

Sports injuries: overuse

with STUART WARDEN

During training, I suffered a stress fracture. It was madness that I carried on and ran the 800 metres and 1500 metres, but because it's the Olympics, you run through the pain.

You never know if you'll have another chance at the Olympics, so unless you can't actually walk you carry on, because you don't want to be asking yourself, 'What if?'

Dame Kelly Holmes, MBE, DBE, 800 and 1500 m gold medallist, Athens 2004 Olympic Games

Overuse injuries present three distinct challenges to the clinician—diagnosis, understanding of why the injury occurred and treatment. Diagnosis requires taking a comprehensive history of the onset, nature and site of the pain along with a thorough assessment of potential risk factors, for example training and technique. Careful examination may reveal which anatomical structure is affected. It is often helpful to ask patients to perform the manoeuvre that produces their pain.

The skilled clinician must seek a cause for every overuse injury. The cause may be quite evident, such as a sudden doubling of training quantity, poor footwear or an obvious biomechanical abnormality, or it may be more subtle, such as running on a cambered surface, muscle imbalance or leg length discrepancy. The causes of overuse injuries are usually divided into extrinsic factors, such as training, surfaces, shoes, equipment and environmental conditions, and intrinsic factors, such as age, gender, malalignment, leg length discrepancy, muscle imbalance, muscle weakness, lack of flexibility and body composition. Possible factors in the development of overuse injuries are shown in Table 4.1.

Treatment of overuse injuries will usually require addressing of the cause as well as specific additional elements such as activity modification, specific exercises to promote tissue repair, soft tissue therapy and pharmacologic agents where appropriate (Chapter 17).

BONE STRESS INJURY

A bone stress injury (BSI) represents the inability of bone to withstand repetitive mechanical loading, which results in structural fatigue and localised bone pain

Table 4.1 Overuse injuries: predisposing factors

Extrinsic factors	Intrinsic factors
Training errors • excessive volume • excessive intensity • rapid increase • sudden change in type • excessive fatigue • inadequate recovery • faulty technique	Malalignment • pes planus • pes cavus • rearfoot varus • tibia vara • genu valgum • genu varum • patella alta • femoral neck anteversion • tibial torsion
Surfaces • hard • soft • cambered	Leg length discrepancy
Shoes • inappropriate • worn out	Muscle imbalance
Equipment • inappropriate	Muscle weakness
Environmental conditions • hot • cold • humid	Lack of flexibility • generalised muscle tightness • focal areas of muscle thickening • restricted joint range of motion
Psychological factors	Sex, size, body composition
Inadequate nutrition	Other • genetic factors • endocrine factors • metabolic conditions

and tenderness. It occurs along a pathology continuum beginning with a stress reaction, which can progress to a stress fracture and ultimately a complete bone fracture.

Pathophysiology

BSIs result from an imbalance between the formation and removal of load-induced damage. A theoretical model is presented in Figure 4.1. Bones deform in response to loading, with the amount of deformation (expressed in terms of strain) depending on the magnitude of load applied and the ability of bone to resist deformation. While the safety factor between exercise-induced strains and those required for complete bone fracture is large, exercise-induced strains can generate microscopic damage (termed 'microdamage').¹ The threshold for microdamage formation depends on the interaction between the number of bone-loading/-strain cycles, the load/strain magnitude and the speed at which load/strain is introduced (strain rate).

Microdamage serves as a stimulus for targeted remodelling.² Targeted remodelling involves activation of remodelling units consisting of advancing front-of-bone-resorbing osteoclasts which remove the damage, followed by rows of bone-forming osteoblasts which deposit layers of new bone. Targeted remodelling maintains homeostasis between damage formation and its removal to preserve bone mechanical properties and enable a bone to adapt

over time to meet changing demands. The adaptation decreases bone strain for a given load,^{3, 4} which means greater loads can be tolerated before the threshold for microdamage formation is surpassed.

Remodelling normally removes damage approximately as fast as it occurs and a reserve exists whereby additional remodelling units can be activated in response to increased microdamage formation. Thus, changes in loading are generally well tolerated. However, remodelling is time dependent and, if insufficient time is given to adapt to a new mechanical stimulus, progressively more damage may form as a result of feedforward between remodelling and damage formation. Feedforward results from resorption preceding formation in remodelling so that an increase in the number of active remodelling units locally reduces bone mass and the energy absorbing capacity of the bone, which potentiates further damage formation.

Accumulating microdamage may combine to initiate the BSI pathology continuum, which includes stress reactions, stress fractures and ultimately complete fracture (Fig. 4.1). Stress reactions are characterised by increased bone turnover associated with periosteal and/or marrow

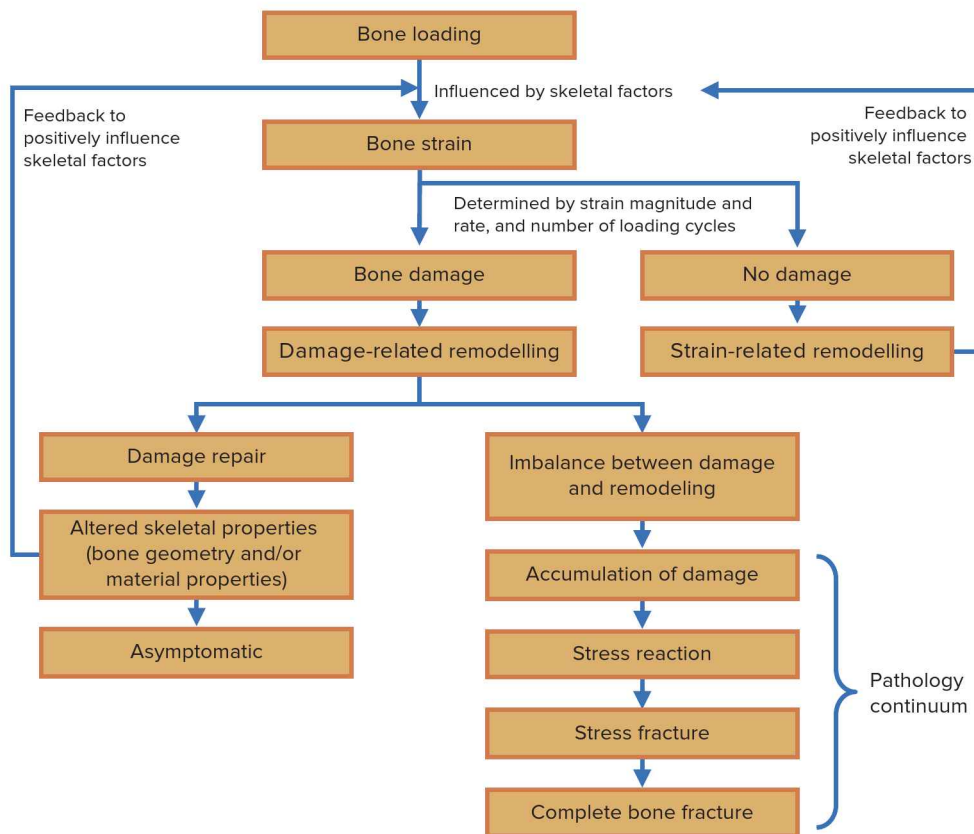


Figure 4.1 A theoretical model of bone stress injury

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oedema, whereas stress fractures have the addition of a discernible fracture line.

Epidemiology

As BSIs develop in response to repetitive mechanical loading, it is not surprising they frequently occur in athletes. In athletes, BSIs cause a considerable loss of participation time and are a significant source of frustration, particularly when they occur in the lead up to a major event or competition. Between one- and two-thirds of runners have a history of a BSI,^{5, 6} while the one-year cumulative incidence of BSIs in cross-country and track and field athletes ranges between 5% and 21%.^{7, 8} Also of note is the high recurrence rate of BSIs, with half of track and field athletes reporting a history of BSIs on more than one occasion and 10–13% of cross-country and track and field athletes with a history of a BSI sustaining a subsequent BSI when followed for 1–2 years.^{6, 7}

BSIs are site specific, occurring at sites exposed to repetitive mechanical loading. Given their association with mechanical loading, BSIs predominantly occur at weight-bearing sites, with the most common sites being the tibia, the metatarsals and the fibula. The exact location depends on how the skeleton is loaded, with different activities resulting in different loading patterns within different

bones. For example, long distance runners typically use a rearfoot strike pattern to preferentially load long bones (tibia, fibula, femur), whereas sprinters use a forefoot strike pattern to introduce relatively greater loads to the bones of the feet. As a result of the differential bone loading, distance runners are at a greater risk of developing long BSIs, whereas sprinters are more prone to BSIs of the tarsals and metatarsals.⁷

Although BSIs predominantly occur at weight-bearing sites, they also occur in response to repetitive loading at non weight-bearing sites.⁹ For instance, rowers repetitively load their rib cage during the drive phase of the rowing stroke and, consequently, are at heightened risk of generating BSIs in their ribs (see Chapter 9).¹⁰ Similarly, throwing athletes are at risk of developing humeral diaphysis BSIs as they repetitively expose their humerus to significant loads during the throwing motion.^{4, 11} As BSIs can occur at both weight-bearing and non weight-bearing sites, they can occur in virtually any bone (Fig. 4.2). Thus, they need to be considered in the differential diagnosis of all presenting injuries.

Risk factors

BSIs can be viewed as occurring when the mechanical stimulus at a specific bone site repetitively exceeds the threshold for microdamage formation. As the bone

Site of stress fracture	Associated sport/activity
Coracoid process of scapula	Trapshooting
Scapula	Running with hand weights
Humerus	Throwing; racquet sports
Olecranon	Throwing; pitching
Ulna	Racquet sports (esp. tennis); gymnastics; volleyball; swimming; softball; wheelchair sports
Ribs—1st	Throwing; pitching
Ribs—2nd–10th	Rowing; kayaking
Pars interarticularis	Gymnastics; ballet; cricket fast bowling; volleyball; springboard diving
Pubic ramus*	Distance running; ballet
Femur—neck	Distance running; jumping; ballet
Femur—shaft	Distance running
Patella	Running; hurdling
Tibia—plateau	Running
Tibia—shaft	Running; ballet
Fibula	Running; aerobics; race-walking; ballet
Medial malleolus*	Running; basketball
Calcaneus	Long-distance military marching
Talus	Pole vaulting
Navicular	Sprinting; middle-distance running; hurdling; long jump; triple jump; football
Metatarsal—general	Running; ballet; marching
Metatarsal—2nd base	Ballet
Metatarsal—5th	Tennis; ballet
Sesamoid bone—foot	Running; ballet; basketball; skating

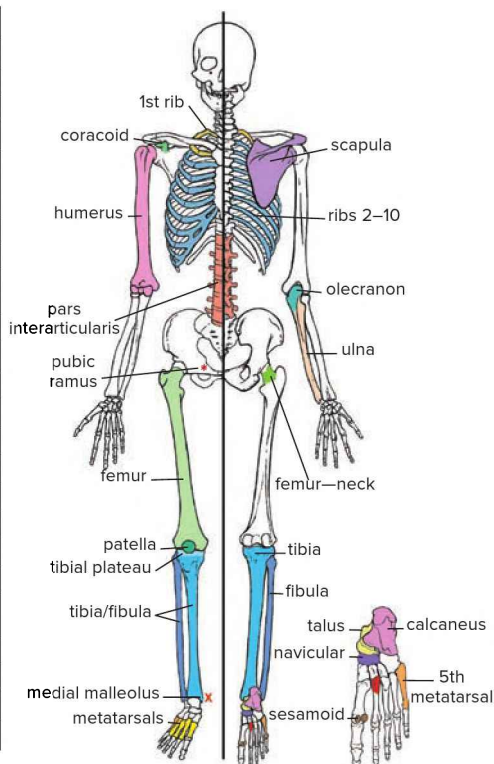


Figure 4.2 Stress fracture site and common associated activity

mechanical stimulus is dependent on the interaction between the applied load and the ability of the bone to resist deformation, risk factors for BSIs can be grouped into two categories:

1. factors modifying the load being applied to a bone
2. factors modifying the ability of a bone to resist load without damage accumulation (Fig. 4.3).

Factors modifying the load applied to a bone

The load applied to a bone during athletic activities represents the summation of external and internal forces, and has magnitude, rate, frequency, duration and direction components. These components influence bone strain magnitude, rate, frequency, duration and location, respectively. Factors modifying the load applied to a bone include biomechanical, training and muscle factors, playing surface and shoes and inserts (orthoses and insoles).

Biomechanical factors

Faulty biomechanics can contribute to BSI risk, and can be dichotomised into those related to abnormal forces and abnormal motions. Increased forces on a normally aligned skeleton can result in abnormal bone loading, with athletes exhibiting high ground reaction force (GRF)

magnitudes and rates, and high accelerations during the early stance phase of running gait being proposed to be at heightened BSI risk. Alternatively, normal forces applied to a malaligned skeleton can also abnormally load bone. Altered movement patterns have the potential to alter the magnitude and/or rate of bone loading, and also alter the direction a bone is loaded and the subsequent distribution of strain within the bone.

The net result may be increased loading of a less accustomed bone site. Statically assessed biomechanical variables that have been implicated in BSI development include external rotation range of motion of the hip,¹³ leg length discrepancy,¹⁴ and both pes planus¹⁵ and cavus.¹⁶ Dynamically assessed variables implicated in BSI development include greater peak hip adduction, knee internal rotation and peak rearfoot eversion in the frontal plane during running¹⁷⁻¹⁹ and less knee flexion in the sagittal plane.²⁰ Having the combination of abnormal forces coupled with a malaligned lower limb is thought to further amplify BSI risk.

Training factors

Chronic introduction of high absolute load magnitudes, rates and accelerations may reduce bone fatigue life, particularly

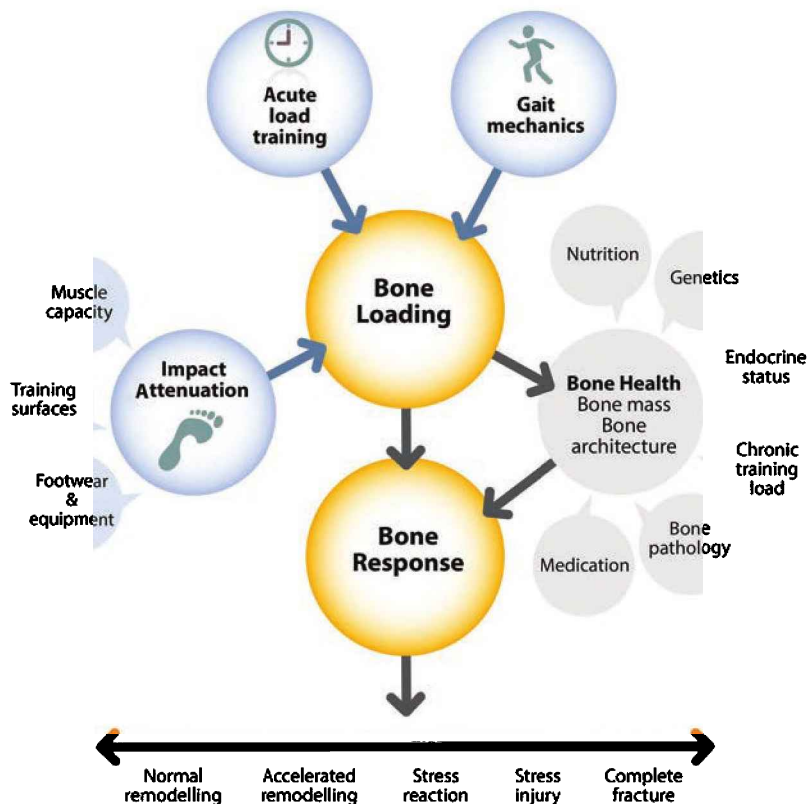


Figure 4.3 Risk factors for bone stress injuries¹²

when the number of loading cycles is high (i.e. running long distances). However, the influence of these variables may be most prominent when an athlete attempts to progress their training. Increases in running speed increase GRFs and their rates of introduction,²¹ while increases in the duration and/or frequency of training sessions increase the total number of bone-loading cycles. In the absence of a change in the load-bearing capacity of a bone, altered loading associated with large changes in training may contribute to microdamage accumulation and the generation of a BSI.

Evidence from military studies confirms that individuals exposed to large changes in physical activity have a heightened BSI risk. For instance, recruits with a lesser history of regular physical activity prior to the commencement of standardised basic training (i.e. those with larger changes in physical activity levels) are at a greater risk of developing a BSI.²²⁻²⁴ While most athletes do not introduce changes in their bone-loading environment to the same extremes as many military recruits, change is a frequent and required means of inducing adaptation. Incrementing training too rapidly or frequently relative to an athlete's usual activities is thought to be central to disrupting the balance between bone microdamage formation and removal.

Muscle factors

Training changes may independently contribute to BSI development, but the relative risk associated with the change may be compounded by muscle factors. An intimate mechanical relationship exists between muscle and bone and the general hypothesis is that muscle is generally protective of BSIs, particularly those occurring in the lower extremity. During impact loading, muscle acts as an active shock attenuator helping to reduce loads as they are transmitted proximally.

When muscles are dysfunctional (weakened, fatigued or altered in their activation patterns) their ability to attenuate loads becomes compromised leading to increased loading of the skeleton. For instance, fatigue results in a decrease in shock attenuation,^{25, 26} increase in GRF loading rates and peak accelerations,^{27, 28} and increase in bone strain magnitude and rate.^{29, 30} In addition, fatigue can lead to altered kinematics, which may alter the direction a bone is loaded resulting in increased bone strain at less accustomed bone sites.³¹ Further support for a protective role of muscle in stress fracture development comes from prospective clinical studies which demonstrated that BSI susceptibility was directly related to muscle size^{14, 32, 33} and strength.³⁴

Playing surface

Playing surface has historically been considered a contributor to BSI risk, with participation on harder surfaces such as asphalt hypothesised to increase loading

compared to participating on softer surfaces such as grass, rubber and sand. However, the interaction between playing surface and injury risk is complex. Athletes alter their leg stiffness when running on surfaces of differing compliance apparently to maintain a constant vertical excursion of their centre of mass.³⁵ Leg stiffness decreases when running on stiffer surfaces to normalise to some extent GRF magnitude; however, GRF loading rates do appear to be systematically increased when running on surfaces that are less compliant.³⁶

Ultimately, what may be important with regards to BSI risk is whether there has been a recent change in running surface that an athlete has yet to become accustomed to. Changes may include increased participation on:

1. less compliant surfaces, which may increase bone strain magnitudes and rate³⁷
2. very compliant surfaces (such as sand), which may increase energy expenditure and influence muscle-related risk factors and kinematics^{38, 39}
3. downhill slopes, which may decrease shock attenuation⁴⁰ and increase loading magnitudes and rates⁴¹
4. altered terrain, which may alter kinematics to load less accustomed skeletal sites.⁴²

Shoes and inserts

The role of shoes and inserts (orthoses and insoles) on bone loading and BSI risk is a topic of ongoing debate. Located at the foot-ground interface, shoes and inserts act as filters that theoretically attenuate ground impact forces. In addition, they have the potential to influence motion of the foot and ankle and the subsequent mechanics proximally in the kinetic chain. Via these two mechanisms, shoes and inserts may influence bone loading and have an effect on BSI risk.

Factors influencing the ability of the bone to resist load without damage accumulation

The amount and rate of strain engendered when a load is applied to a bone is dependent upon the ability of the bone to resist deformation in the direction of loading. For a given applied load, less rigid bones experience greater strain and at a faster rate than more rigid bones and, thus, are more susceptible to microdamage and BSI formation. Skeletal features that influence bone rigidity include the amount of bone material present (mass) and its distribution (structure). There is strong evidence that both contribute to BSI risk^{14, 23, 33, 43, 44} and therefore it is important to consider modifiable factors that may contribute to these skeletal characteristics. Three modifiable factors in athletes that may impact the ability of bone to resist loading and contribute to BSI risk are physical activity history, energy availability, and calcium and vitamin D status.

Physical activity history

A longer history of physical activity appears to be protective against the development of a BSI.^{22-24, 45, 46} An improved ability of the skeleton to resist loading likely contributes to the reduced BSI risk in individuals with a prior history of physical activity. The skeleton responds and adapts to mechanical loading in a site-specific manner to increase its rigidity in the direction of loading. It principally does this during the growing years by preferentially depositing small amounts of new bone on the outer periosteal surface at a distance from the bending axes. As the rigidity of a unit area of bone is proportional to the fourth power of its distance from a bending axis, the addition of a small amount of mass to the outer surface of a bone results in a disproportionate increase in bone strength. The net result is a decrease in the bone strain engendered in response to a given load and increase in bone fatigue life.³

Energy availability

Gender factors contribute to BSI susceptibility, with females being at greater risk.⁴⁷ The cause for the higher incidence of BSIs in females appears to relate to the interrelationships between energy availability, menstrual function and bone mass—otherwise known as the female athlete triad. While athletes can have one or more components of the triad, low energy availability appears to be the central factor.⁴⁸ Low energy availability results from insufficient dietary intake to meet exercise energy expenditure. It can result from low dietary energy intake (whether inadvertent, intentional or psychopathological) and/or excessive exercise energy expenditure. The menstrual and skeletal changes associated with low energy availability reduce the ability of bone to resist load and/or impair its ability to repair microdamage. The net result is heightened BSI risk, particularly in elite female long distance runners where the difference between dietary energy intake and exercise energy expenditure is often small and the incidence of menstrual dysfunction is elevated.

Calcium and vitamin D status

Calcium combines with phosphate to form hydroxyapatite crystals to endow bone with rigidity, while vitamin D contributes by promoting calcium absorption in the gut and reabsorption in the kidneys. Prospective studies provide evidence for roles of both low calcium and vitamin D in BSIs,^{49, 50} and a large randomised controlled trial demonstrated a 20% reduction in BSI incidence in female Navy recruits with suboptimal baseline daily calcium intake when they were supplemented with daily calcium and vitamin D.⁵¹ Overall, the data suggest that athletes should ensure sufficient calcium and vitamin D

intakes to meet or exceed the current recommended dietary allowances of 1000–1300 mg and 600 IU (for individuals aged 14–50 years), respectively.⁵²

Diagnosis

BSIs often present with a history consistent with an overuse injury—the gradual onset of activity-related pain. As BSIs occur along a pathology continuum, signs and symptoms may vary depending upon what point along the continuum the athlete presents. An astute clinician may be able to diagnose the pathology at the stress reaction stage. However, some patients may not present until the pathology has progressed to a stress fracture where there is actual cortical disruption. The earlier within the continuum an athlete presents and a diagnosis is made, the more likely the pathology may respond quickly and favourably to management. Thus, bone as the tissue of origin of an athlete's symptoms needs to be considered at all times during differential diagnosis to ensure prompt diagnosis and management.

The diagnostic features of a stress fracture are shown in Table 4.2.

History

A thorough history is the first step to the diagnosis of a BSI. In most instances, individuals with a BSI have a consistent and predictable history that centres on pain. At the start of the pathology continuum, pain is usually described as a mild diffuse ache that occurs after a specific amount of activity and at specific times (depending on which bone is afflicted and when loaded during activity). The pain does not tend to resolve or 'warm-up' as activity is continued and only abates once activity (i.e. bone loading) is ceased.

As the initial pain often subsides soon after activity completion and is not present during rest it is often ignored at first. However, with continued training and progression of the pathology the pain may become more severe and localised and occur at an earlier stage. It may also persist for longer periods following the completion of activity and begin to be present during activities that involve lower levels of bone loading, such as walking. Eventually, the pain can result in activity restriction or the

Table 4.2 Diagnostic features of a stress fracture

- Localised pain and tenderness over the fracture site.
- A history of a recent change in training or taking up a new activity.
- X-ray appearance often normal or there may be a periosteal reaction (Fig. 4.4).
- Abnormal appearance on MRI (Fig. 4.5), radioisotopic bone scan (scintigraphy) (Fig. 4.7) or CT scan (Fig. 4.8).

need to cease training altogether. At this more advanced stage, any associated inflammatory response to the injury may also occasionally contribute to resting and night pain.

Examination

On physical examination, the most obvious feature of a BSI is localised bony tenderness. Certain bones (such as the tibia, fibula and metatarsals) lend themselves well to palpation because of their well-defined anatomical boundaries and the absence of overlying muscle. In these relatively subcutaneous bones, precise and thorough palpation is required as tenderness may be localised with adjacent areas being completely pain-free. Occasionally redness, swelling and warmth from an associated inflammatory reaction may also be felt, along with periosteal thickening and callus formation in longer-standing BSIs.

Direct palpation is obviously not possible at deeper sites (such as the femur and pars interarticularis of the spine), with symptoms at these sites possibly being provoked by specific bone-loading tests, such as hopping,⁵³ the fulcrum test for the femoral shaft,⁵⁴ or the one-legged hyperextension test for the pars interarticularis.⁵⁵ However, the sensitivity and specificity of these bone-loading tests have either not been investigated or disputed.⁵⁶ Likewise, the application of a vibrating tuning fork or therapeutic ultrasound for the clinical diagnosis of BSIs is not supported.⁵⁷

Imaging

Imaging plays a significant role in the diagnosis. Imaging options include plain radiography, bone scan, computed tomography (CT), magnetic resonance imaging (MRI) and, more recently, ultrasound.

Plain radiographs have poor sensitivity and may not detect stress injury until the injury has developed along the bone stress continuum.^{58, 59} Although some old textbooks suggested that stress fractures become visible on plain radiographs after 2-6 weeks, prospective studies prove that some stress fractures remain invisible on plain radiography. Radiographic changes, when present, include subtle focal periosteal bone formation (Fig. 4.4), or, later, frank cortical defects.

In countries where physicians have ready access to MRI, this is generally the first line investigation of bone stress injuries. MRI is sensitive in detecting pathophysiological changes in soft tissue, bone and marrow associated with bone related stress injuries. It can reveal abnormalities in these structures before plain radiographic changes and has comparable sensitivity to a bone scan.⁵⁸ Other advantages of MRI for bone imaging are its multiplanar capability (which helps the clinician precisely define the location and extent of injury), lack of exposure to ionising radiation and significantly less imaging time than a three-phase bone scan.



Figure 4.4 X-ray showing periosteal new bone formation indicative of a stress fracture

The typical MRI appearance of a stress fracture shows periosteal and marrow oedema plus or minus the actual fracture line (Fig. 4.5). Several fat-suppression techniques such as short tau inversion recovery are used to maximise the sensitivity of MRI in bone stress injuries.⁶⁰

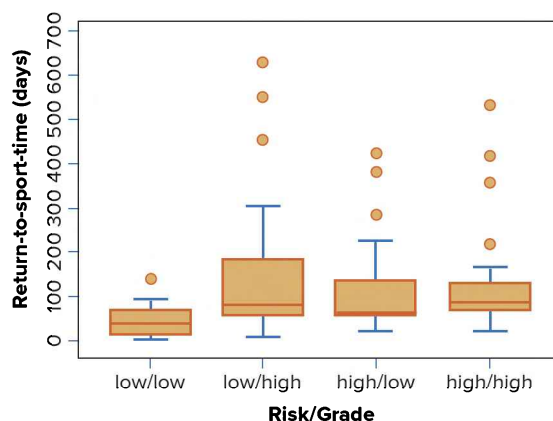
Various grading systems for stress fractures based on MRI appearances have been suggested.⁶¹ The most commonly used systems⁶²⁻⁶⁴ are shown in Table 4.3.

Attempts have been made to use the MRI appearance to predict return to play (RTP). In a prospective study involving 34 fractures among 211 college athletes, MRI grading severity, lower bone mineral density (BMD) and the fracture site (trabecular > cortical bone) were important variables associated with time to return to sport.⁶⁴ The mean time to full return to sport for grade 4 BSIs (31.7 +/- 3.7 weeks) was significantly higher compared with grade 3 (18.8 +/- 2.9 weeks), grade 2 (13.5 +/- 2.1 weeks) and grade 1 injuries (11.4 +/- 4.5 weeks). Among grade 3 and 4 injuries, the mean time to return to sport was significantly longer for those at sites high in trabecular versus cortical bone (38.1 +/- 6.4 vs 18.8 +/- 2.1 weeks, respectively).⁶⁴



Figure 4.5 MRI of a stress fracture showing bony oedema (white)

It should be recognised that the above values are based on the average time taken to return to sport. However, as illustrated in Figure 4.6, this fails to reflect the wide variation among individuals—particularly those with higher MRI-grade injuries or injuries located in high-risk sites (covered in the section ‘Classification of bone stress injuries’).⁶⁵ Care should therefore be taken in making definitive return-to-sport predictions based on imaging alone, except perhaps in the case of mild stress reactions in low-risk sites.



(Dots indicate outliers).

Figure 4.6 Return-to-sport times for BSIs grouped according to a combination of their location (low risk/high risk) and MRI grade (low: grades 1 and 2; high: grades 3 and 4)⁶⁶

Table 4.3 MRI grading scales for BSIs

Grade	Fredericson et al. ⁶²	Arendt et al. ⁶³	Nattiv et al. ⁶⁴
1	Mild to moderate periosteal oedema on T2 Normal marrow on T2 and T1	Positive signal change on STIR	Mild marrow or periosteal oedema on T2 T1 normal*
2	Moderate to severe periosteal oedema on T2 Marrow oedema on T2 but not T1	Positive STIR plus positive T2	Moderate marrow or periosteal oedema plus positive T2 T1 normal
3	Moderate to severe periosteal oedema on T2 Marrow oedema on T2 and T1	Positive STIR plus positive T2 and T1	Severe marrow or periosteal oedema on T2 and T1
4	Moderate to severe periosteal oedema on T2 Marrow oedema on T2 and T1 Fracture line present	Fracture line present on T2 or T1	Severe marrow or periosteal oedema on T2 and T1 Fracture line present on T2 or T1

STIR: short T1 Inversion recovery

*Radiograph results are often negative at all grades; they may be normal, or a periosteal reaction may be evident

Beck and colleagues in their study included grading systems for radiography, isotope bone scan, CT scan, as well as MRI and found that only the MRI grading system had any relationship to clinical severity.⁶⁷

A radioisotopic bone scan was the most important diagnostic test for athletic stress fractures in the 1990s.⁶⁰ The technical aspects of this modality are outlined in Chapter 15. The appearance of a BSI on a bone scan is a focal area of increased uptake (Fig. 4.7).

Note that bone scans lack specificity for stress fractures—bony abnormalities such as tumours, especially osteoid osteoma and osteomyelitis, have similar appearances. It may also be difficult to precisely localise the site of the area of increased uptake. Increased uptake can occur in non-painful sites, indicating subclinical accelerated remodelling. As with plain film, CT and MR imaging, imaging appearance returns to normal after clinical resolution because of ongoing bony remodelling.⁶⁰

CT is less sensitive than a bone scan or MRI in the early detection of a BSI.^{58, 59} However, it is more sensitive than both radiographs and MRI for the detection of cortical fracture lines. CT is thus well suited to demonstrate stress fractures of the sacrum, pars interarticularis, tarsal navicular and longitudinal stress fractures of the tibia.⁵⁸

A CT scan will clearly image the fracture (Fig. 4.8) and differentiate between a stress fracture (positive bone scan, clear fracture line) and a stress reaction (positive bone scan and negative CT scan). CT can also distinguish a BSI and other causes of hot bone scans such as osteoid osteoma and osteomyelitis.

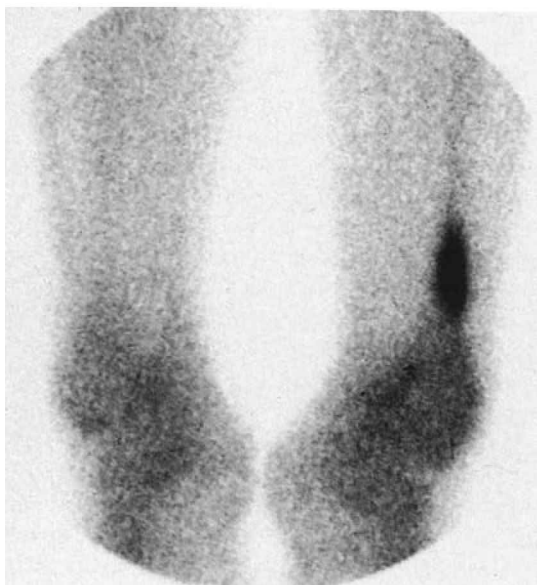


Figure 4.7 Radioisotopic bone scan of a stress fracture
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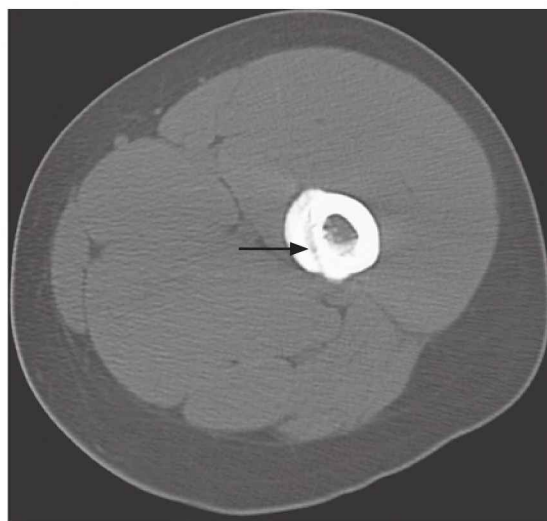


Figure 4.8 CT scan of a stress fracture showing a cortical defect (arrowed)

Ultrasound has some potential in the diagnosis of stress fractures but it is not ready for routine clinical use.⁵⁹

Ultimately, athletes who display clinical signs and symptoms of a possible BSI require imaging to confirm suspicions and make a definitive diagnosis. Briefly, in most clinical settings, plain radiographs remain the first line of imaging for BSIs because of their low cost and wide availability; however, radiographs are limited by their planar nature and low spatial resolution, which contribute to extremely low sensitivity. CT also lacks sensitivity, but may be utilised in specific cases where demonstration of a fracture line may affect treatment. In contrast, bone scintigraphy has high sensitivity, but is limited by low specificity and the use of extremely high ionising radiation doses. Of the imaging modalities currently available, MRI is the modality of choice because of its superior contrast resolution, lack of exposure to ionising radiation, and its combined high sensitivity and specificity.

The different imaging appearances of the various stages of the BSI continuum are shown in Table 4.4.

Classification of bone stress injuries

BSIs can be dichotomised into either low or high-risk groups according to their location (Table 4.5).⁶⁸ Low-risk BSIs predominantly occur on the compression side of a bone's bending axis, and have a favourable natural history in that recovery occurs with a low incidence of complications and without the need for aggressive intervention (such as surgery and/or prolonged modified weight-bearing). In contrast, high-risk BSIs often occur on the tension side of a bone's bending axis, and present treatment challenges demanding specific attention

Table 4.4 Imaging findings with different phases of the BSI continuum

Clinical features	Early bone stress	Stress reaction	Stress fracture
Local pain	Nil	Usually	Always
Local tenderness	Nil	Usually, if superficial	Usually, if superficial
X-ray appearance	Normal	Normal	May be abnormal (periosteal reaction or cortical defect in cortical bone, sclerosis in trabecular bone)
Radioisotopic bone scan appearance	Increased uptake	Increased uptake	Increased uptake
CT scan appearance	Normal	Normal	Features of stress fracture (as for X-ray)
MRI appearance	May show increased high signal	Increased high signal	Increased high signal $\pm$ cortical defect

Table 4.5 Low-risk and high-risk BSIs

Low risk	High risk
Posteromedial tibia	Femoral neck
Fibula/lateral malleolus	Anterior cortex of the tibia
Femoral shaft	Medial malleolus
Pelvis	Talus (lateral process)
Calcaneus	Navicular
Diaphysis of second to fourth metatarsals	Proximal diaphysis of the fifth metatarsal
	Base of second metatarsal
	Great-toe sesamoids

because they are prone to delayed or non-union and/or are at high risk for progression to complete fracture.

Management

This section focuses on general management principles for a low-risk BSI, such as one occurring at the posteromedial border of the tibia. The management of specific high-risk BSIs is summarised in Table 4.6 and detailed management is included in chapters covering different regions of the body.

Low-risk BSIs represent relatively straightforward management problems in the sense that they readily heal without complication. A two-phase approach consisting of modified activity followed by a gradual resumption of activity forms the cornerstone of management. While the overarching goal is to return the athlete to his or her

Table 4.6 Stress fractures that require specific treatment other than rest

Stress fracture	Treatment
Femoral neck	Non-displaced: initial bed rest for 1 week, then gradual weight-bearing Displaced: surgical fixation
Anterior tibial cortex	Non weight-bearing on crutches for 6–8 weeks or intramedullary screw fixation
Medial malleolus	Non weight-bearing cast immobilisation for 6 weeks or surgical fixation
Talus (lateral process)	Non weight-bearing cast immobilisation for 6 weeks or surgical excision of fragment
Navicular	Non weight-bearing cast immobilisation for 6–8 weeks or surgery
Metatarsal—2 nd base	Non weight-bearing for 2 weeks; partial weight-bearing for 2 weeks
Metatarsal—5 th base	Cast immobilisation or percutaneous screw fixation
Sesamoid bone of the foot	Non weight-bearing for 4 weeks

pre-injury level of function in the shortest time possible without compromising tissue level healing, it is generally acknowledged that there is more to the successful management of athletes with a BSI. The high recurrence rate of BSIs indicates that a central goal for clinicians when managing low-risk BSIs should be to identify and modify potential risk factors for future BSIs. Using the concept that BSIs occur when the load being applied exceeds a bone's ability to resist the load without microdamage accumulation, runners with a BSI require strategies to either reduce the load being introduced or increase the load-bearing capacity of their skeleton.

Phase 1. Initial management

Activity modification

There is no question that temporary discontinuation of usual activity and the introduction of a variable period of modified activity is required in the initial management of low-risk BSIs in order to permit tissue-level healing and prevent pathology progression. However, the duration and extent of activity modification is highly variable and is decided upon on an individual basis using pain as the principal guiding variable. The presence of pain either during or after an activity indicates that the pathological site is being excessively loaded for the current stage of healing and that loading needs to be titrated.

In the initial stages of BSI management, the goal is for the athlete to be pain-free during and after usual activities of daily living (ADLs). Cushioned shoes and/or insoles may provide assistance with dissipating impact forces during ADLs in athletes with a rearfoot or leg BSI,^{67, 68} whereas athletes with a BSI within the mid or forefoot may consider wearing a stiff soled shoe to reduce bending forces and symptoms.⁶⁹

Walking should be minimised to that essential to perform ADLs and must use a normal gait pattern. If a normal gait pattern cannot be used or symptoms are produced either during or after walking, partial weight-bearing via the use of assistive gait devices should be considered. Alternatively, a pneumatic leg brace may be introduced to promote pain-free gait in the case of BSIs of the fibula or posteromedial border of the tibia.⁷⁰ If a pain-free normal gait pattern cannot be achieved, a short period in a walking boot or of non weight-bearing may be considered. However, in each scenario progression to unassisted pain-free gait should be sought as soon as possible.

Athletes need to be pain-free not only during, but also after activities. The presence of resting and/or night pain is a sign that the underlying pathology may have an inflammatory component. While resting pain does not typically require specific intervention as it is usually short term and abates with activity modification, some

athletes may consider non-steroidal anti-inflammatory drugs (NSAIDs) or other analgesics. The use of these agents should be discouraged beyond a few days because of their ability to mask pain and subsequently influence activity progression and their potential to impede tissue-level healing when taken for prolonged periods.⁷¹

Identification and initial management of potential risk factors

The initial period following BSI diagnosis is a useful time to evaluate and begin addressing potential contributing factors as it is often the time when an athlete gives most attention. A detailed activity history is important. Consider not only usual and recent changes in activity frequency, duration and intensity, but also unusual or new participation in physical activities beyond running. BSI risk reflects the sum of all bone loading and loading from non-running activities may be enough to push an athlete beyond their injury threshold. Note should also be taken of any recent changes in running surfaces, shoes/ inserts or technique. By combining knowledge of recent running progressions with knowledge of lifelong physical activity history, it may be possible to provide a runner returning from a BSI with advice with regards to future running program design. For instance, novice runners with a minimal physical activity history may need to progress their training program at a slower rate so as to not overload the skeleton and disrupt the homeostasis between microdamage formation and removal.

History taking in an injured athlete should also assess factors influencing the ability of their skeleton to resist loading. It is essential to obtain a history of the patient's general health, medications and personal habits to ensure that there are no factors that may influence bone health. A past history of BSIs (and other bone injuries) and current body mass index of less than 19 are strong risk factors for BSIs that require assessment and thorough exploration.^{6, 8}

A full dietary history should be taken with particular attention paid towards the possible presence of deficits in energy intake and/or eating disorders, and calcium and vitamin D intake. A maternal family history of osteoporosis or low bone mineral density should be explored and a detailed menstrual history should be taken in female runners including their age of menarche and subsequent menstrual status. Identification of any issues of concern warrants appropriate referral when indicated, remembering that consultation for a BSI may be the first time that an athlete's issues associated with their relative energy deficiency are identified.

In addition to history taking, initial assessment and management of potential factors influencing bone loading can be explored. To guide the interventions and formulate

an initial biomechanical impression of the athlete, it is important to gather as much circumstantial evidence as possible. Such evidence may come from recent videos of the athlete running, assessment of their shoe wear pattern, their history of overuse injuries and whether there is a unilateral predominance, static posture and alignment, and hip, knee and foot mechanics during non-painful activities such as walking or single-leg squats.

Depending on the individual athlete and the preliminary examination findings, initial biomechanical interventions can be introduced to maintain what the athlete already has or address suspected deficits. Activities can include muscle strength, endurance and control training and interventions for muscle length and joint mobility. Key areas to consider include control, endurance and strength at the hip, knee and ankle, core stability, and strength of the intrinsic and extrinsic muscles of the foot. The activities may need to be modified in order to be pain-free, and are often performed in non or partial weight-bearing.

Maintenance of physical conditioning

Maintaining conditioning during recovery is important for as seamless a return to activity as the athlete's pathology permits. Conditioning activities should be introduced early and can include cycling, swimming, deep-water running and anti-gravity treadmill training. The latter two methods may be most specific in running athletes as they more closely reproduce the neuromuscular recruitment patterns involved in running. Anti-gravity treadmill training is not introduced until a runner is pain-free during both walking and ADLs. In the interim, deep-water running can be introduced as long as the athlete is pain-free both during and after sessions.

Accelerating tissue healing

It would be beneficial to provide an athlete with a means of accelerating healing of their BSI. While there are no proven methods of accelerating BSI healing, a number of candidate methods have been proposed. Low intensity ($<1.0 \text{ W/cm}^2$) pulsed ultrasound (LIPUS) therapy currently has the greatest support given its observed beneficial effects on complete bone fractures.⁷² LIPUS has been effective in stimulating union in 98% and 94% of BSIs displaying delayed and non-union, respectively.⁷³ However, the benefits of LIPUS on uncomplicated BSIs currently remain unclear. Extracorporeal shock wave therapy and electromagnetic and capacitive coupled electric fields have also been considered as modalities for promoting BSI repair, but studies of their effects have principally been limited to recalcitrant BSIs. At this stage, the use of electrotherapeutic agents to accelerate BSI healing cannot be recommended.

A number of pharmaceutical agents may facilitate BSI healing but currently none are accepted as standard treatment. They include bisphosphonates⁷⁴ and the anabolic therapies parathyroid hormone and anti-sclerostin antibody therapy. Parathyroid hormone, when administered intermittently, promotes osteoblastogenesis and osteoblast survival. Anti-sclerostin antibody therapy inhibits osteocyte-secretion of sclerostin to facilitate Wnt signalling and subsequently promote osteoblast proliferation and function. Both agents stimulate bone formation and accelerate fracture healing in preclinical models. Whether the same fracture healing benefits carry over to humans and to BSIs currently remains unknown.⁷⁵

Phase 2. Return to running

Beginning and progressing initial running

A graduated running program can be used in low-risk BSIs to introduce controlled loading and facilitate a return to activity in a timely, yet safe manner. While loading is central to the development of low-risk BSIs, recovery is best met by a balance between rest from aggravating activities and performance of appropriate loading. Appropriate loading can be defined as loading that does not provoke BSI symptoms either during or after completion of an activity. Once an athlete with a low-risk BSI becomes pain-free during unassisted walking, they can start reintroducing running-related loads. While there is no established protocol for returning to running during recovery from a low-risk BSI, a prescriptive program to facilitate the return of a recreational athlete to 30 minutes of running is provided in Table 4.7. The program consists of a pre-entry stage and three running stages.

Once the athlete is completely pain-free for five consecutive days during usual activities, they leave the pre-entry stage (stage 0) and commence deliberate progressive loading. Stage 1 introduces loading in 30-minute sessions separated by rest days. Sessions in this stage consist of increasing durations of jogging (defined as running at 50% of normal pace) and decreasing durations of walking. The pace of running is progressed in stage 2 until the athlete can run for 30 minutes at their usual pace, with stage 3 consisting of running on two consecutive days followed by a rest day. The last stage incorporates individualised running until there is complete return to desired running activities.

Progress through each stage of the graduated running program is determined by BSI provocation. If an athlete is able to complete a session with no BSI symptoms and they do not experience latent symptoms, they can progress to the next level during the ensuing session. However, if BSI symptoms are experienced during or after a session, the athlete must stop the session and at the next session return to the last level they were able to successfully complete.

Table 4.7 Graduated running program to return a runner to 30 minutes of pain-free running¹²

Stage 0 Pre-entry to graduated running program		
		Pain during walking in normal ADLs
Stage 1 Initial loading and jogging (50% normal pace) with increasing duration		
Level	A	Walk 30 minutes
	B	Rest
	C	Walk 9 minutes and jog 1 minute (x3)
	D	Rest
	E	Walk 8 minutes and jog 2 minutes (x3)
	F	Rest
	G	Walk 7 minutes and jog 3 minutes (x3)
	H	Rest
	I	Walk 6 minutes and jog 4 minutes (x3)
	J	Rest
	K	Walk 4 minutes and jog 6 minutes (x3)
	L	Rest
	M	Walk 2 minutes and jog 8 minutes (x3)
	N	Rest
Stage 2 Running with increasing intensity		
Level	A	Jog 30 minutes
	B	Rest
	C	Run 30 minutes at 60% normal pace
	D	Rest
	E	Run 30 minutes at 60% normal pace
	F	Rest
	G	Run 30 minutes at 70% normal pace
	H	Rest
	I	Run 30 minutes at 80% normal pace
	J	Rest
	K	Run 30 minutes at 90% normal pace
	L	Rest
	M	Run 30 minutes at full pace
	N	Rest
Stage 3 Running on consecutive days		
Level	A	Run 30 minutes at full pace
	B	Run 30 minutes at full pace
	C	Rest
	D	Run 30 minutes at full pace
	E	Run 30 minutes at full pace
	F	Rest
	G	Run 30 minutes at full pace
Stage 4 Return to running		

Using a graduated loading program that is strictly guided by pain, the athlete and clinician can facilitate recovery while being relatively confident that there is no pathology progression. This requires the full understanding of the athlete regarding the appropriate progression through the program and adherence to the set pain levels. Patients with lower grade BSIs may be able to progress more quickly through a graduated loading program and the total duration of their program may be reduced from that outlined. Program duration should be based on the expected time course for recovery and, thus, all patients should perform individualised programs.

Anti-gravity treadmill training

When available, an anti-gravity treadmill may initially be used in the place of overground running as anti-gravity treadmill training may allow an athlete to run at higher intensities earlier during recovery, but with lower bone loading. Thereby, anti-gravity treadmill training can be used to maintain fitness while protecting the BSI site. Anti-gravity treadmills consist of a treadmill with an air-filled pressure-controlled chamber that surrounds the lower half of the body from the waist down. Pressure in the chamber is modulated to unweight the runner in 1% increments so that they are running with between 100% and 20% body weight.

Runners with a low-risk BSI can start anti-gravity treadmill training once they are pain-free during walking and ADLs for at least a week. A typical starting point is to jog every other day three times for five minutes with between 50% and 70% body weight and one-minute recovery between repetitions. This is performed for an initial week to acclimatise to anti-gravity treadmill training while symptoms are monitored for provocation. How to progress from this initial stage is currently somewhat of an art rather than a science; however, a proposed progression is, over the next 2 weeks, to increase the duration of each run to a total of 20–30 minutes, the body weight percentage by 5–10% and the running speed.⁷⁶ The progression continues thereafter as long as there are no BSI symptoms until the athlete is running at their usual speed and duration with 90% body weight. Once this is achieved, ground running is introduced and relatively quickly progressed over a couple of weeks to normal durations and intensities.

Running gait retraining

Despite a slow, progressive return to a running program, the persistence of faulty mechanics may hinder the healing process or contribute to elevated risk for a repeat injury. Therefore, it is important to identify and address any underlying mechanics when treating a runner with a BSI. Using the hypothesis that bone loading is directly

related to GRF and acceleration parameters, interventions that reduce GRFs and shock during running gait may represent a means of decreasing BSI risk. By reducing the magnitude and rate of bone loading, the number of loading cycles until microdamage accumulation and fatigue failure may be increased. A number of gait retraining techniques are currently being investigated to reduce bone loading during running, including:

- the use of accelerometer-based biofeedback to encourage reductions in loading magnitude and rate^{77, 78}
- increasing stride rate to reduce stride length and, subsequently, vertical excursion and velocity of the centre of mass, and GRFs and tibial accelerations⁷⁹⁻⁸¹
- modifying initial contact to encourage a more forefoot, rather than heel or rearfoot, strike pattern.^{82, 83}

These gait retraining techniques should not be taken lightly. Athletes have typically been running with their particular gait pattern for many years and the ability of an athlete's musculoskeletal system to adapt to the nuances of their particular gait should not be underestimated. However, a history of repeat BSIs and the accumulating loss of running time is a sign that gait retraining should be considered. In doing so, it must be remembered that, when inducing a change in gait, there is always the potential of altering injury risk at an alternative site. Thus, transitioning to a new running gait should be performed slowly and may be benefited by a preconditioning program.

Running program design

Incrementing a training program too rapidly or frequently relative to an athlete's usual activities can contribute to the generation of a BSI by upsetting the balance between bone microdamage formation and its removal. Unfortunately, there is no accepted algorithm for how much an individual can modify their training program before excessively heightening their BSI risk. The historical rule of thumb has been to change a training program by no more than 10% per week to reduce injury risk. However, this 'rule' is not generalisable on an individual basis, with different runners being able to tolerate more or less change before development of an injury.⁸⁴

In order to provide an athlete returning from a BSI with appropriate advice regarding the design and advancement of their running program, it is important that they recall as much as they can of their pre-injury training regime. This should include information on changes in any feature that may have altered the load being introduced to the skeleton, including training intensity, duration, frequency, type, surface, technique, shoes and so on. In addition, recovering athletes should be encouraged to maintain a training diary containing these

data, to not only track their training progress, but also to provide reliable data regarding their training program for future reference.

In addition to the obvious limiting of the magnitude and number of training variables changed at any one given time, the use of cyclic training methods should be encouraged. Cyclic training may involve the introduction of rest periods into a training program, or the replacement of overground training sessions with lower bone-loading activities such as deep-water running or anti-gravity treadmill training. This may involve a monthly regime of 3 weeks of training and 1 week of rest or low-load activities.

Healing of bone stress injuries

Healing is assessed clinically by the absence of local tenderness and functionally by the ability to perform the precipitating activity without pain. It is not useful to attempt to monitor healing with X-ray or radioisotopic bone scan.⁸⁵ CT scan appearances of healing stress fractures can be deceptive as in some cases the fracture is still visible well after clinical healing has occurred.⁸⁶ Figure 4.9 shows the relationship between imaging appearance and the various stages of bone stress.

Osteitis and periostitis

Osteitis (impaction trauma or primary inflammation of bone) and periostitis (abnormal histological appearance of periosteal collagen) are also considered overuse injuries. The condition known as osteitis pubis occurs in the pubic bones of the pelvis and is characterised by deep-seated pain and tenderness of the symphysis pubis with generalised increased uptake on the radioisotopic bone

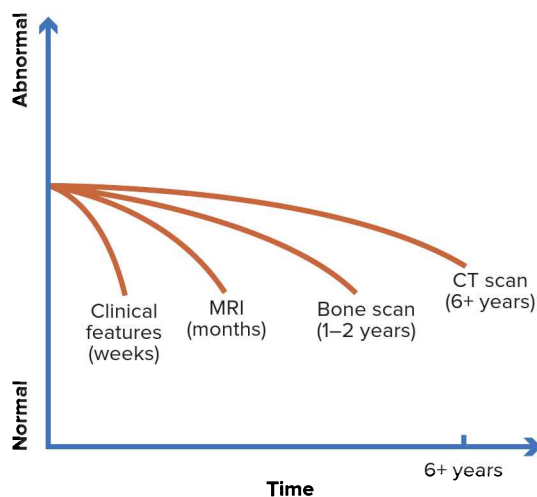


Figure 4.9 MRI, bone scan and CT scan return to their normal appearance well after clinical union occurs

scan. The exact pathogenesis of this injury remains in debate (Chapter 32).

Periostitis occurs commonly, mainly at the medial border of the tibia, a condition often known as 'shin splints'. In this condition, tenderness along the medial border of the tibia at the interface between the muscles and their bony attachments corresponds with an area of increased uptake on the bone scan.

The treatment of periostitis consists of local symptomatic therapy as well as unloading the muscle contraction on the periosteum. In the shin, strain may be reduced by altering the biomechanics through controlling excessive pronation. Soft tissue therapy and stretching may also be effective.

Apophysitis

Bony injury may occur at the attachment of the strong, large tendons to the growth areas. This condition is called 'apophysitis' and the most common examples are Osgood-Schlatter disease at the attachment of the patellar tendon to the tibial tuberosity and Sever's disease at the attachment of the Achilles tendon to the calcaneus. A full description of apophysitis is given in Chapter 44.

ARTICULAR CARTILAGE

Overuse injury can affect the articular cartilage lining of joints, particularly in osteoarthritis. Changes range from microscopic inflammatory changes to softening, fibrillation, fissuring and, ultimately, to gross visible changes. In younger people, this pathology can arise at the patella (patellofemoral syndrome), but it is important to note that the pain of patellofemoral syndrome can occur in the presence of normal joint surfaces. This very common condition is discussed in Chapter 36.

JOINT

Inflammatory changes in joints associated with overuse are classified as synovitis or capsulitis. Examples of these problems are the sinus tarsi syndrome of the subtalar joint and synovitis of the hip joint.

Impingement syndromes occur when a bony abnormality, either congenital or acquired, causes two bony surfaces to impinge on each other (e.g. femoro-acetabular impingement at hip, posterior impingement at ankle) or impinge on a structure passing between them (e.g. supraspinatus tendon in shoulder) causing damage to that structure. Treatment requires either removal of the structural abnormality or modification of biomechanics to relieve the impingement.

LIGAMENT

Overuse injuries of ligaments are uncommon, but may occur as a result of excessive load.

Injury to the medial ulnar collateral ligament of the elbow in baseball pitchers is the most common ligamentous overuse injury.⁸⁷

MUSCLE

Muscle may respond to overuse in a number of ways, including myofascial pain, chronic compartment syndromes and exercise-induced muscle soreness.

Myofascial pain: trigger points or just sore spots of unknown origin?

The clinical phenomenon of localised palpatory muscle pain is one that is well known to sports medicine practitioners. The concept of myofascial pain caused by trigger points is a popular explanatory model with several putative pathophysiological mechanisms having been proposed to explain these findings.

The trigger point model

The myofascial trigger point theory was pioneered by Travell and Simons over 30 years ago.⁸⁸ The theory postulated by Travell and Simons has, until recent times, had widespread acceptance. Trigger points are classified as either active or latent, with only active trigger points being symptomatic. Active trigger points are thought to be tender to palpation, generate a local twitch response on stimulation and create a predictable pattern of referred pain on compression.

The term myofascial pain syndrome is commonly used to describe the clinical signs thought to represent active trigger points. As per Travell's original description, this is characterised by:

1. localised areas of deep muscle tenderness or hyperirritability, referred to as 'trigger points'—palpable taut bands within the muscle associated with muscle inhibition, intolerance to stretch, as well as autonomic symptoms such as vasoconstriction or dilation
2. a predictable, discrete reference zone of deep aching pain which is worsened by palpation of the trigger point. This may be located in the immediate region of or remote from, the trigger point.

Dommerholt expanded on Travell's theory, postulating a role for excessive release of acetylcholine from dysfunctional neuromuscular endplates, causing taut bands and subsequent muscle ischaemia. This muscle ischaemia was thought to precipitate an energy crisis in the muscle, causing the release of pro-inflammatory molecules, thereby activating nociceptive neurons.⁸⁹ Newer theories currently subject to research include the principle of neurogenic inflammation leading to sensitisation of innervated muscle,⁹⁰ secondary allodynia

secondary to irritation of deeper structures such as ligament or bone (central sensitisation),⁹¹ descending inhibitory pain modulation,⁹² and diffuse noxious inhibitory control.⁹³

Despite the abundance of hypothesised theories regarding the pathophysiology of myofascial trigger points, their existence is not widely accepted in the medical literature. Furthermore, much of the research to date is limited by significant methodological flaws.⁹⁰ Consequently, despite many years of research endeavour, the diagnosis, treatment and pathophysiological mechanisms underlying localised palpatory muscle pain remains an area of vigorous debate.^{90, 94} Despite this, the lack of definitive anatomical or biochemical evidence has not prevented the concept of trigger points from achieving popularity among sports and musculoskeletal medicine clinicians.

While many alternative hypotheses of myofascial trigger points have been proposed since Travell's original description, none have been widely accepted. Consequently, there are no generally accepted diagnostic criteria for myofascial trigger points.⁹⁵ Meakins articulated a key component of this controversy by coining the term 'sore spots of unknown origin'.⁹² This reflects the unreliability of identifying taut bands in skeletal muscle⁹⁶⁻⁹⁹ while acknowledging the acceptable reliability of localising muscle tenderness based on patient feedback, presumably through the removal of the subjective phenomenon of palpatory pareidolia.¹⁰⁰

A diagnosis of trigger points causing pain must only be made following the appropriate exclusion of alternative diagnoses. Failure to respond to appropriate treatment should necessitate a review of the diagnosis.

Pain from trigger points

The onset of pain from trigger points may follow either an acute or subacute pattern. Some patients may be able to recall the onset of pain, often noting a precipitating event, while others will report a gradual onset of pain over time. Where a precipitating event is recalled, the initial amplitude of pain is often only minor, with a subsequent progressive increase over time.

In contrast, the onset of pain due to acute mechanical tissue trauma, as with a ruptured ankle ligament, displays a very rapid onset to near maximal severity (though exceptions such as meniscal tears do exist). Subacute onset trigger point pain may be more difficult to identify on history, as patients often present following many months of pain. In this situation, identification of aetiological or exacerbating factors, such as inappropriate loads or sustained postures, is useful.

Fluctuations in pain, over the course of a day or longer periods, are typical of trigger points and suggestive that

acute tissue trauma is not involved. Fluctuations usually follow particular events, such as lying on a particular side in bed or computer-based work. This is quite distinct from the diurnal variation as seen in inflammatory conditions. While the pain of trigger points has a different temporal pattern to that of tissue trauma, it is important to note that trigger points are often secondary to and therefore frequently coexist with acute tissue trauma.

The spatial distribution of pain also differs between that of trigger points and localised tissue trauma. Travell describes predictable referred pain patterns attributed to myofascial trigger points,⁹⁷ derived from the injection of hypertonic saline into muscle. As referred pain tends to be poorly localised, patients often indicate the location of referred pain with a wave of the hand. Direct stimulation of tissue nociceptors in contrast, as may occur with a ligament rupture, typically results in more precisely located pain, able to be indicated with a pointed finger.

Clinical features

Recent years have seen attempts to utilise medical imaging techniques in the detection of myofascial trigger points. Unique features of localised elliptical hypoechogenicity corresponding to palpable findings have been demonstrated with sonoelastography,⁹⁸⁻¹⁰⁰ while magnetic resonance elastography has demonstrated the ability to identify increased tissue stiffness in muscle.^{101, 102} A lack of control groups in this research, however, limits the ability to draw firm conclusions regarding the role of myofascial trigger points in these findings.

While Travell describes palpable taut bands in muscle as a diagnostic feature of trigger points,⁹⁷ several controlled studies have found this finding to be unreliable and therefore not recommended.¹⁰³⁻¹⁰⁶

In contrast, the identification of localised muscle tenderness based on patient feedback has shown acceptable inter-examiner reliability,⁹⁶ presumably by removing the subjective phenomenon of palpatory pareidolia. Thus, on the basis of existing research, the clinical identification of trigger points should be based primarily on localisation of maximally tender points within muscle. Consideration also may be given to the reproduction of any referred pain patterns.

Treatment

Evidence for the effectiveness of trigger point treatments is inconclusive.¹⁰⁷ Despite widespread anecdotal reports of effectiveness, systematic reviews and blinded randomised placebo-controlled studies show no clinical worthwhile effect of these treatments over sham, standard care or exercise therapy.¹⁰⁸⁻¹¹⁰ The apparent contradiction between the anecdotal experience of many clinicians and research findings may be explained by the phenomenon

of the localised twitch response, an involuntary localised transient contraction within the muscle being treated. Many experienced practitioners elicit multiple twitch responses when treating a muscle. In contrast, much of the research conducted on trigger point therapy has not placed any emphasis on the localised twitch response, failing to account for its presence or absence in findings of efficacy.

A variety of methods of treating trigger points have been described in the literature, with the most effective method unclear. These include spray and stretch, digital ischaemic pressuring, dry needling, wet needling, proprioceptive neuromuscular facilitation and muscular 'stripping'. Published research on these methods is frequently sparse, confounded by methodological limitations or both.

The most rigorously researched of the above methods are dry needling, wet needling and digital ischaemic pressure. Wet needling is generally considered to be at least equal to alternative methods. In many jurisdictions, however, injection therapy is limited to appropriately qualified medical practitioners. The application of dry needling has been shown to be a reasonable alternative. Subsequent digital ischaemic pressure applied immediately following needling therapy may further improve efficacy.

Dry needling and other techniques used in the treatment of trigger points are discussed in Chapter 17.

Chronic compartment syndrome

Chronic compartment syndrome (CCS) refers to the intermittent and reversible pathologic elevation of compartment pressures following exertion.¹¹¹ The condition usually affects the lower leg but may also occur in the forearm in the sports of tennis, rock climbing and weightlifting.¹¹²

The muscles of the lower leg are divided into a number of compartments by fascial sheaths, which are relatively inelastic thickenings of collagenous tissue. Exercise raises the intracompartmental pressure and may cause local muscle swelling and accumulation of fluid in the interstitial spaces. The tight fascia prevents expansion. This impairs the blood supply and causes pain with exertion. Compression of neurological structures may also contribute to the clinical presentation. Muscle hypertrophy may also precipitate CCS.

The main symptom of CCS is pain that commences during activity and ceases with rest. This differs from other overuse injuries such as tendinopathies, where pain may be present with initial exercise, then diminish as the affected area warms up, only to return following cessation of activity. The role of imaging in the diagnosis of CCS is unclear.¹¹³

The gold standard for diagnosis of CCS has always been measurement of compartment pressures at rest and during

pain-provoking exercise. Compartment pressure testing is described in Chapter 15. In recent years, various authors have expressed reservations regarding the reliability and validity of these tests.¹¹⁴⁻¹¹⁶

Surgical treatment—fasciotomy (release of the fascia) or fasciectomy (removal of the fascia)—has always been the mainstay of treatment of compartment syndromes. Results of surgery are often disappointing with a high incidence of recurrence of exercise-related leg pain. The role of surgery is now being questioned.¹¹⁷

A biomechanical basis of the exercise-related leg pain has been suggested by Franklyn-Miller,¹¹⁸ and changes in running gait has shown promising initial results in the management of these patients.¹¹⁹

Exercise-induced muscle soreness

Exercise-induced muscle soreness is a common complaint during or immediately following vigorous unaccustomed exercise. Muscle soreness that develops 24–48 hours after unaccustomed high-intensity activity is known as delayed onset muscle soreness (DOMS). DOMS appears to be more severe after activities that involve eccentric muscle contractions; that is, when muscle contracts whilst simultaneously lengthening. Activities that involve both high muscle forces and eccentric muscle contractions, such as skiing and downhill running, are a common cause of DOMS in untrained individuals.

The exact cause of acute exercise-induced muscle soreness and DOMS is unclear. Although debate in the literature about the mechanism of muscle soreness is ongoing, the general consensus is that the soreness is caused by microtrauma of muscle cells and connective tissue. Microtrauma is followed by a local inflammatory process within the extracellular space which sensitises nerve endings via mechanical, chemical or thermal stimulation.¹²⁰⁻¹²²

Signs and symptoms

Exercise-induced muscle soreness is often accompanied by muscle stiffness, ache and mild cramping. Symptoms can persist for several hours post exercise before resolving.¹²¹ DOMS is characterised by tenderness on palpation and a sense of stiffness with physical activity. Soreness can peak around 48 hours after exercise. Resolution of symptoms can take up to 10 days.¹²⁰ Soreness can vary from mild discomfort to incapacitating pain; this is largely dependent on the intensity and volume of training, as well as the actual inducing event.

Importantly, DOMS can be associated with muscle strength deficits, which may affect physical performance. Strength deficits associated with DOMS are typically greatest between 24 and 48 hours after activity.^{123, 124} A reduction in muscle power and reduced joint range

of motion may also affect an athlete's performance. Local muscle swelling and intracellular proteins in the blood are also common signs.^{120, 123-125} Exercise-induced muscle soreness and DOMS are often associated with elevated levels of creatine kinase in the blood. However, given the self-limiting nature of acute muscle soreness and DOMS, blood tests are rarely warranted and the clinical interpretation of such tests is unclear.¹²⁵

Management

Management strategies in the treatment of DOMS focus on encouraging healing and recovery of the inflamed and damaged muscle tissue. Intense exercise during the symptomatic period can potentially cause more micro-trauma and muscle injury, particularly as strength, power and coordination of movement are compromised. Therefore, during the symptomatic period, training and competition performance should be modified accordingly.

Common modalities used to treat exercise-induced muscle soreness and DOMS include massage, cryotherapy, stretching and active recovery. The management of DOMS is described in Chapter 13.

Prevention of exercise-induced muscle soreness and delayed onset muscle soreness

Prevention of exercise-induced muscle soreness should be centred on a properly planned training program, which adequately addresses strength, power, endurance and quality of movement. Adequate periods of training and conditioning are needed before individuals participate in strenuous activity to ensure the exercise-induced muscle soreness is minimised and more serious injuries are prevented. DOMS occurs less in those who train regularly, although even trained individuals may become sore after an unaccustomed exercise bout. There is little evidence to support the use of static stretching to prevent acute muscle soreness and DOMS.¹²¹ Variable results have been shown with preventative approaches such as the use of vitamins C and E and protein supplements.¹²⁶ Understanding an athlete's current training volume including exercise intensity, duration, rate of progression and recovery will provide a context for understanding and preventing future episodes of muscle soreness.

TENDON

Tendon overuse injury (tendinopathy)

with JILL COOK, CRAIG PURDAM

Tendon overuse injuries provide a major proportion of the sports clinician's workload. The clinical presentation is straightforward in many cases—the patient presents with tendon pain during or after activity (Table 4.8)

and loading tests demonstrate increased pain with increased load.

In the 1980s, the underlying pathology was often referred to as 'tendinitis'—this was associated with a belief that cellular inflammation contributed to the pathological process. Histopathological studies in the 1990s indicated that inflammatory cells were absent in patients who underwent surgery for tendon pain. Recently, low levels of inflammatory cells and markers have been demonstrated, but it is unlikely that inflammation is a key driver of tendon pathology and a resident cell-driven process remains the best-supported pathoaetiology.

The pathological findings at surgery are consistent with 'tendinosis'. The boxed item illustrates tendon pathology—'tendinosis'—at the anatomical level (Fig. 4.10) and at two microscopic levels (Figs. 4.11 and 4.12). The pathology is also summarised in Table 4.9.

A contemporary model of a continuum of tendon pathology

Tendon authorities Jill Cook and Craig Purdam have proposed that tendon pathology should be considered as a continuum.¹²⁷ They contend that the dichotomy of 'normal' and 'degenerative tendinosis' is too simplistic. Importantly, their three-part classification has implications for treatment, and so it is summarised here and illustrated (Fig. 4.13).

Stage 1. Reactive tendinopathy

This refers to the non-inflammatory response of tendon cells and matrix proteins to an acute tensile or compressive overload. Tendon cells become activated and may proliferate in this stage—they become more prominent—consistent with the appearance of cells that

Table 4.8 Clinical presentation of patients with overuse tendon pain (tendinosis)

- Pain after exercise or, more frequently, the following morning upon rising.
- It can be pain-free at rest and initially becomes more painful with use.
- Athletes can 'run through' the pain or the pain disappears when they warm up, only to return after exercise when they cool down.
- Athletes are able to continue to train fully in the early stages of the condition; this may interfere with the healing process.
- Examination reveals local tenderness and/or thickening.
- Frank swelling and crepitus may be present, although crepitus is more usually a sign of associated tenosynovitis (it is not 'inflammatory fluid').

Tendinosis: what is it?

This box illustrates the pathology found at end-stage tendinopathy—when symptoms have been present for at least 3 months. The illustration (Fig. 4.10) is based on pathological specimens (Figs 4.11 and 4.12) obtained at surgery for chronic sports-related tendon pain.

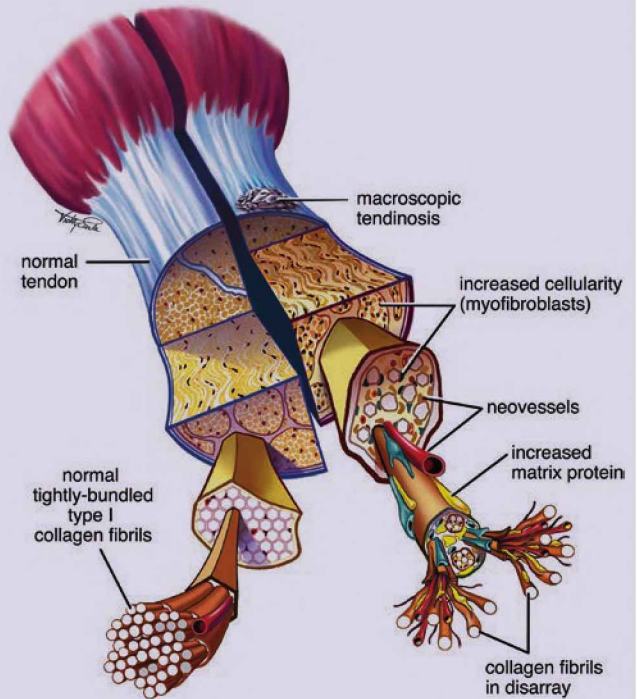
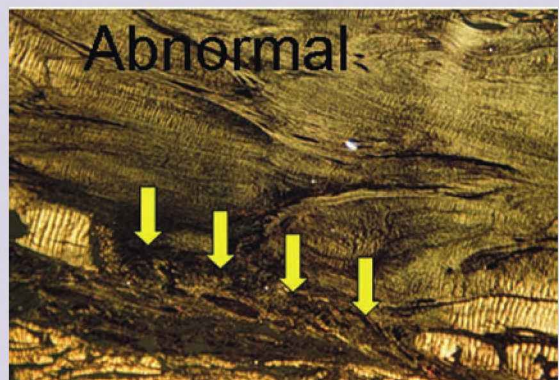


Figure 4.10 The contrasting features of normal tendon (left side) and tendinosis (right side). Characteristic features at this macroscopic level are the collagen fibres of different sizes in disarray, abnormal cell numbers (decreased and increased), abnormally prominent blood vessels and an increase in matrix proteins



(a)



(b)

Figure 4.11 Under polarised light microscopy: (a) normal tendon has tightly bundled parallel collagen fibrils with a characteristic golden reflectivity; (b) a specimen from a patient with chronic patellar tendinopathy showing collagen fibril separation and frank discontinuity within some fibrils

continued

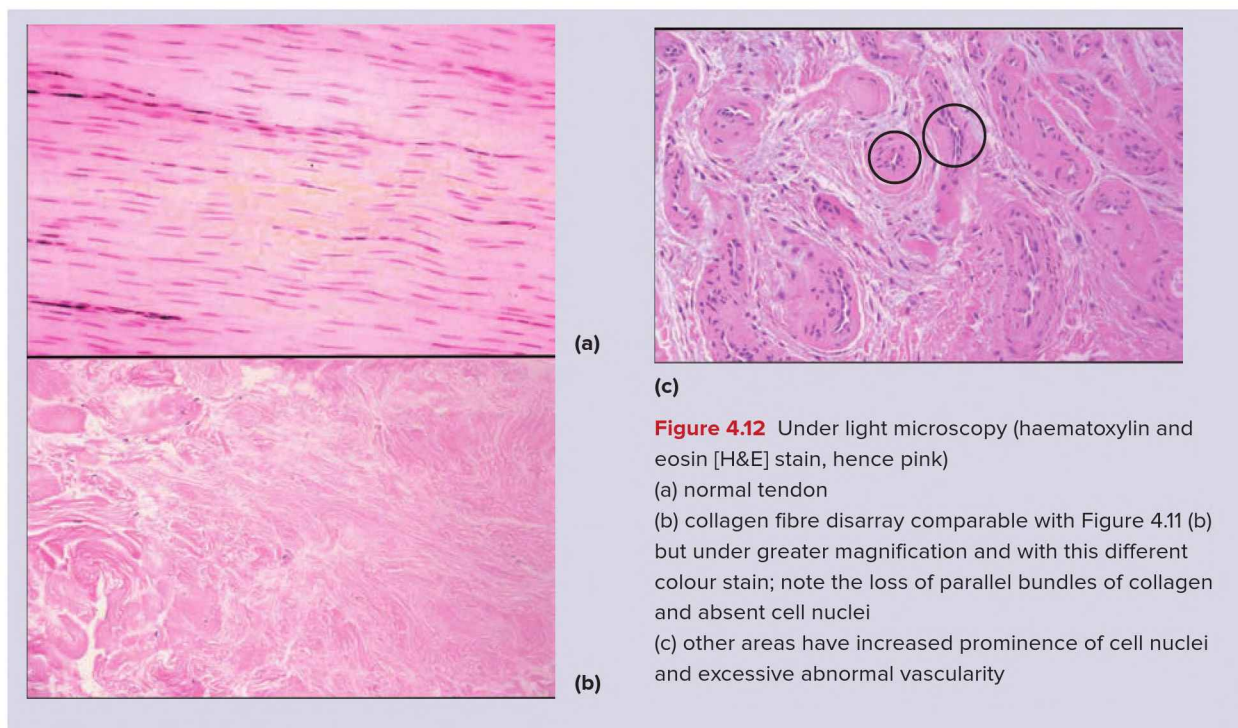


Table 4.9 Five elements of normal tendon compared with the characteristic elements of end-stage tendon overuse injury

Tendon element	Normal tendon	Changes that occur in response to excessive tendon loading
Cells—tenocytes	Tendon cells are spindle-shaped and nuclei cluster in longitudinal chains on microscopy	Tissue has proliferation of cells with abnormally rounded nuclei (Fig. 4.12c) and areas with fewer than normal cell numbers (Fig. 4.12b)
Ground substance or 'matrix' proteins	The ground substance in the matrix is minimal and is not visible when stained for light microscopic viewing	Increased amount of ground substance/matrix proteins which stain and are visible under light microscopy
Collagen	Linear and tightly bundled and has a characteristic crimp under polarised light	Disrupted—both longitudinally and in its bundles (Fig. 4.12b)
Nerves	Minimal intratendinous nerves, some innervation of connective tissue in and around the tendon	Abnormal ingrowth of nerves (mostly sympathetic) and a preponderance of neuropeptides
Vessels	Minimal vascularity when examined histologically or by using ultrasound	Prominent vessels histologically or using ultrasound (Fig. 4.12c)

are producing repair proteins, especially proteoglycans. This results in a short-term thickening of a portion of the tendon that reduces stress. This differs from the normal tendon adaptation to tensile load that generally occurs through tendon stiffening with little change in thickness. This is seen clinically in the acutely overloaded tendon and is more common in the younger

person. It also arises when there is trauma to a tendon, such as a direct blow.

At this stage, both ultrasound and MRI show mild fusiform swelling—greater tendon diameter with little disruption of the collagen fascicles. The change in imaging appearance is mainly derived from the increase in bound water within the matrix proteins (proteoglycans).

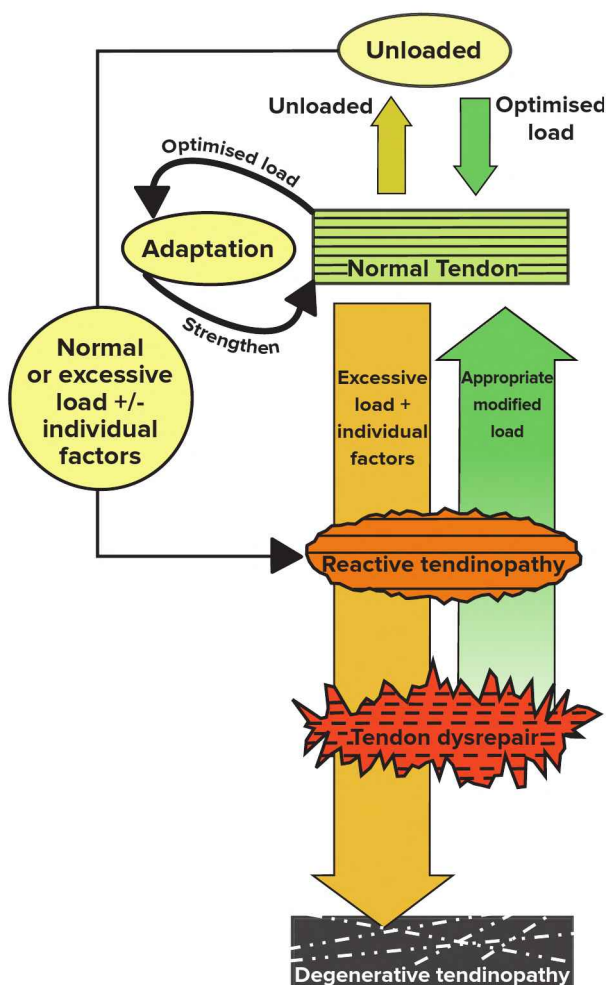


Figure 4.13 The Cook–Purdam model to help clinicians understand the relationship between loading/unloading and the several stages of tendon pathology

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Stage 2. Tendon dysrepair

This describes a worsening tendon pathology with greater matrix breakdown. Tendon cells are more prominent and take on a rounded appearance (chondrocytic). Protein production increases—both matrix proteoglycans and collagen. As a result of these changes, collagen separates and the matrix becomes somewhat disorganised. The disruption of the matrix may allow for some ingrowth of vessels and nerves.

This is seen clinically in overloaded tendons in the young, but it may appear across a spectrum of ages and loading environments. The transition from the previous stage may be difficult to detect clinically, but imaging

will reveal more focal changes of hypoechogenicity on ultrasound. There may be a mild increase in vascularity on colour or Doppler ultrasound. MRI reveals a swollen tendon with increased signal.

Stage 3. Degenerative tendinopathy

This is the stage that is present in patients who undergo surgery for chronic tendon pain—it is the ‘end stage’ of tendon overuse injury. The matrix and cell changes described in stages 1 and 2 may progress so that areas of apoptosis (absent cell nuclei due to cell death) are evident. Large areas of matrix are disordered and filled with cells, vessels, matrix breakdown products and disordered type I, II and III collagen. The structure of the tendon is heterogeneous—degenerative pathology is interspersed between other stages of pathology and normal tendon.

This pathology exists in the older person, or the younger person or elite athlete with long-term tendon overload. Typically, it is seen in a middle-aged, recreational athlete with focal tendon swelling and pain (e.g. mid-Achilles region). The tendon may have focal nodular areas with or without general thickening. This is the type of tendon that may rupture if the degenerative pathology is extensive.

In this stage, compromised matrix and vascular changes are obvious on ultrasound scans as hypoechogenic regions with few reflections from collagen fascicles. Larger vessels are usually prominent on Doppler ultrasound. MRI shows increased tendon size and intratendinous signal. The changes are more focal than spread throughout the tendon.

Reactive on degenerative tendon

Degenerative tendinopathy exists in a tendon that has normal tissue. When these tendons are overloaded, the degenerative part of the tendon is unlikely to take load because of a lack of matrix structure, so the load is borne by the normal part of the tendon. Overload can induce a reactive change in this normal tendon, resulting in a reactive on degenerative tendon.

Other terms associated with overuse tendon injuries

Although the most-used clinical label for tendon overuse injuries is ‘tendinopathy’ as above and is used in specific chapters of this book (i.e. Achilles tendinopathy), the terms ‘paratenonitis’, ‘partial tear’ and ‘tendinitis’ need definition.

Paratenonitis

This term includes peritendinitis, tenosynovitis (single layer of areolar tissue covering the tendon) and tenovaginitis (double-layered tendon sheath). This

occurs in situations where the tendon rubs over a bony prominence and/or where repeated movement directly irritates the paratenon. Uncommonly, it can coexist with partial tears and tendinosis.

Partial tears

The term 'partial tear of a tendon' should be reserved for a macroscopically evident subcutaneous partial tear of a tendon, an uncommon acute, not overuse, injury, at least in the Achilles and patellar tendon (see Chapter 3). The distinction on imaging between tendon degeneration and partial tear is difficult because there are no definite signs that allow a clear distinction and diagnosis is reliant on radiologist expertise.

Principles of rehabilitating lower limb tendinopathy

Before starting

Ensure that the diagnosis is correct and the pain is from the tendon. Tendon pain will respond best to the loading program described.

Decide the start and end of the program

Start at the functional level and capacity of the person. This can be determined from a comprehensive examination of function and its relationship to pain. The program should end at the functional level required for the person to return to their desired activity level. This can be determined from their history; both the sport that the person has been involved in as well as the sport or activity and level they want to return to. The sport and the level will determine how much endurance is required for the tendon, associated muscle and kinetic chain.

A schematic of tendon rehabilitation, improving tendon capacity with progressive loads, is shown in Figure 4.14.¹²⁸

Stages of rehabilitation

Stage 1. Isometric exercise

Isometrics are a great way to relieve tendon pain, hence a good spot to start as most clinical presentations are for tendon pain. Isometric exercises result in pain relief in

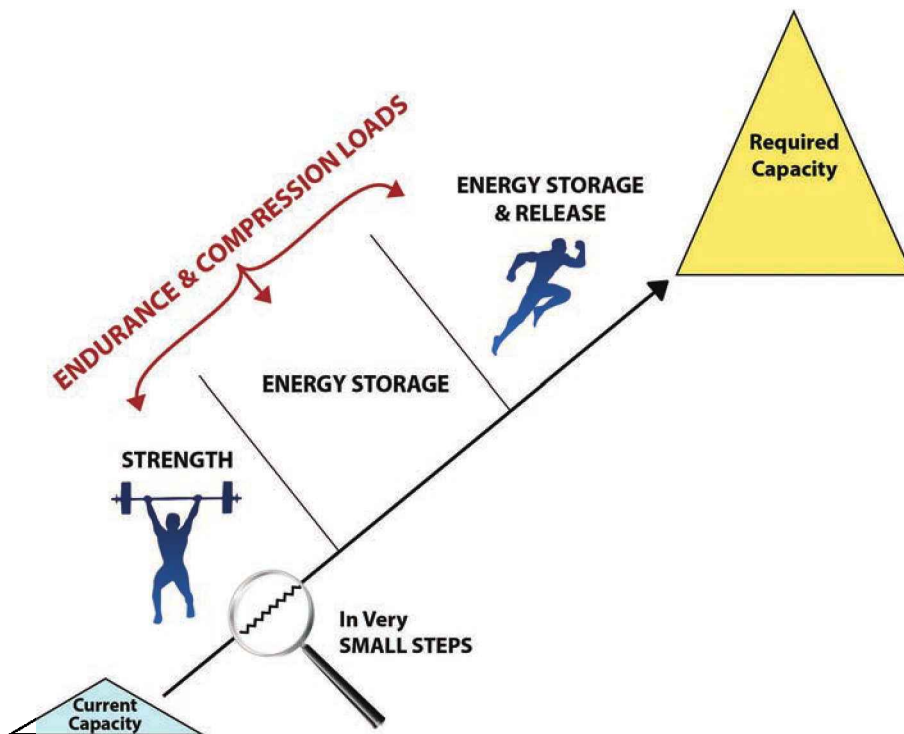


Figure 4.14 Schematic of tendon rehabilitation, improving tendon capacity with progressive loads. Introduction and progression of endurance and compressive loads are critical within each stage. The start and end points of rehabilitation will vary between individuals

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7-10 days in a reactive on degenerative tendon, but can take up to 6-8 weeks in a purely reactive tendon.

Isometrics have been shown to immediately relieve patellar tendon pain more than isotonic exercise and the pain relief was sustained for 45 minutes.¹²⁹ Rio et al. also showed in this study that isometric exercise normalised cortical inhibition that was excessive in patellar tendinopathy, whereas isotonic exercise had no effect on cortical inhibition.¹²⁹

Clinically, isometrics are therefore used to manage pain in the reactive or reactive on degenerative phase of tendon pathology. The research protocol in young, jumping men was four repetitions of 45 second holds at 70% of maximal voluntary contraction and this level of loading is appropriate for the painful athletic tendon.¹²⁹ This level of loading may be too much for the older person with clinical tendinopathy, and varying both the load and time held is necessary. Using body weight loads for longer holds is a simple place to start and adding weights often requires equipment. As isometrics may need to be done more than once a day, doing them at home is the best option. Hiring gym equipment such as a seated calf raise for a few months can help adherence to regular isometric exercise.

Stage 2. Isotonic exercise

Once the pain has settled and/or is stable at a low level, isotonic exercises can be started. These are slow and heavy concentric/eccentric exercises directed at improving strength in the muscle tendon unit. These are best performed with a slightly longer eccentric component (e.g. three-second concentric, four-second eccentric) for brain plasticity. These should be performed for every muscle in the kinetic chain that has a functional deficit—most likely the anti-gravity muscles such as the calf (gastrocnemius and soleus), quadriceps and gluteal muscles. Hamstrings are clearly addressed as a priority in hamstring tendinopathy.

Isotonic exercises should be continued until there is good strength in the affected musculotendinous unit and the agonists in kinetic chain function. This can take up to 12 weeks in a profoundly unloaded limb. An endurance set is added if the person requires higher levels of endurance for their activity. Endurance can also be added with weight-bearing exercise, such as stair climbing for either Achilles or patellar tendinopathy.

PRACTICE PEARL

Performing a bout of isometrics before isotonic exercises increases maximum voluntary contraction, most likely due to cortical pain inhibition.¹²⁹

Stage 3. Energy storage exercise

Once the muscles are stronger, the tendon can be loaded with energy storage exercises. To store energy in a tendon you need to do faster eccentric exercises, initially with a slower release (concentric phase), but building to stage 4 with quick energy storage and faster release. As these exercises place high loads on the tendon they are best completed only 2-3 times a week. These faster exercises need to be controlled in terms of numbers and/or time, started at a low level and gradually increased.

PRACTICE PEARL

It is essential to continue isotonic exercises at least two days a week throughout the entire rehabilitation program, as well as after the athlete has returned to full training and competition. Continued isometric loading before isotonic exercises may also be helpful throughout the whole program.

Stage 4. Sport-specific energy storage and release exercise

Once the tendon can tolerate high-energy storage loads, then functional activity-specific exercises can be introduced in a clinical setting. These will replace stage 3 exercises and can only be completed 2-3 times a week. Once the tendon is tolerant to training numbers of high-level functional activities then the person can return to a controlled training environment and from there to a graduated return to competition.

Enthesopathy

Enthesopathies occur at the bone-tendon junction, such as the Achilles tendon insertion onto the calcaneus. Enthesopathies are similar to tendinopathies in that they are primarily degenerative in nature. Treatment principles are the same.

BURSA

The body contains many bursae situated usually between bony surfaces and overlying tendons. Their role is to facilitate movement of the tendon over the bony surface. All bursae are susceptible to injury. Typically, injuries to bursae are overuse injuries resulting from excessive shearing and/or compressive forces.^{130, 131}

Overuse injuries in bursae are quite common, particularly at the subacromial bursa,¹³² the greater trochanteric bursa,¹³³ the bursa deep to the iliotibial band at the knee,^{131, 133} and the retrocalcaneal bursa separating the Achilles tendon from the calcaneus.¹³⁴ Overuse

pathologies affecting bursae commonly couple with other local pathologies such as tendinopathies and impingement syndromes.^{135, 136} Symptoms include localised pain and swelling and typically increase with activity. Conservative approaches to treatment are often trialled first.

Treatment involves removal of irritating loads, reduction of inflammation and a progressive return to pain-free activity. Specific treatments including ice, electrical stimulation, iontophoresis and gentle stretching. Once initial inflammation subsides, a stretching and strengthening program for the surrounding tissues and muscles can commence. NSAIDs are widely prescribed for these conditions and corticosteroid injections, often guided by ultrasound, are considered where conservative approaches have failed.

NERVE

Nerve entrapment syndromes occur in athletes as a result of swelling in the surrounding soft tissues or anatomical abnormalities. These may affect the suprascapular nerve, the posterior interosseous, ulnar and median nerves in the forearm, the obturator nerve in the groin, the posterior tibial nerve at the tarsal tunnel on the medial aspect of the ankle and, most commonly, the interdigital nerves, especially between the third and fourth toes, a condition known as Morton's neuroma. This condition is not a true neuroma but rather a nerve compression. These nerve entrapments occasionally require surgical decompression.

Chronic mild irritation of a nerve may result in damage manifested by an increase in neural mechanosensitivity. These may be the primary cause of the patient's symptoms or may contribute to symptoms. This concept is discussed more fully in Chapters 5 and 6.

SKIN

The skin's integrity is constantly challenged by athletic activity, weather conditions and pathogenic organisms.

Blisters

The skin of many athletes is subjected to friction-related forces and compounded by perspiration.^{137, 138} Exposure to shearing and compressive forces can lead to mechanical separation of the epidermal cell layers. Hydrostatic pressure causes further separation and allows plasma-like fluid or sweat into the space to form a blister.¹³⁷ The repair process starts 24 hours post incident and blisters generally heal in approximately five days.¹³⁷ Blisters may occur at any site of friction with an external source, such as shoes or sporting equipment.

Blisters are common in marathon runners, rowers, race walkers, triathletes, hikers and in military populations. Foot blisters are painful and can have an impact on sporting performance. Sock type, race, previous hiking

or military experience and certain orthopaedic foot conditions predict the development of foot blisters in the military population.¹³⁸

Foot blisters can be prevented by wearing-in new shoes, wearing socks and smearing petroleum jelly over the sock at sites of friction. Strategies to prevent blisters also serve to prevent callus. Symptomatic callus can be pared down with a scalpel blade, taking care not to lacerate the normal skin.

At the first sign of a blister, the aggravating source should be removed and either adhesive tape applied over the blistered area or blister pads should be applied. Blister pads prevent blisters by acting as a barrier between skin and shoe.

Treatment of blisters involves prevention of infection by the use of antiseptics and protection with sticking plaster. Fluid-filled blisters may be punctured and drained.

Skin infections

Almost any cutaneous infection can afflict athletes. However, their activities place these individuals at higher risk to develop and subsequently transmit their skin ailment to competitors. Athletes acquire infections as a result of their interaction with other athletes and with the environment in which they compete. An athlete's skin is often macerated from sweating which promotes common infections in sports including bacterial, fungal and viral infections, but parasites can also afflict the athlete.¹³⁹

Dermatitis

An athlete's skin suffers repeated exposure to trauma, heat, moisture and numerous allergens and chemicals. These factors combine with other unique and less well-defined genetically predisposing factors in the athlete's skin to cause both allergic contact dermatitis (ACD) and irritant contact dermatitis (ICD). As with other cases of contact dermatitis, these eruptions in athletes present as a spectrum of acute to subacute to chronic dermatitis. Recognising the unique environmental irritants and allergens encountered by athletes is paramount to facilitate appropriate therapy and prevention.¹⁴⁰

Skin cancers

Skin cancers are not true overuse injuries, but perhaps can be described as 'overexposure.' Nevertheless, athletes are particularly vulnerable as many sports involve prolonged exposure to sun during training and games.

The incidence of melanoma and non-melanoma skin cancers is increasing worldwide. Ultraviolet light exposure is the most important risk factor for cutaneous melanoma and non-melanoma skin cancers. Non-melanoma skin cancer includes basal cell carcinoma and squamous cell carcinoma. Constitutive skin colour and genetic factors,

as well as immunological factors, play a role in the development of skin cancer. Ultraviolet light also causes sunburn and photoaging damage to the skin. Studies conducted within athletic populations have found that certain sports are associated with increased risk for skin cancers.¹⁴¹

Athletes should take care to avoid excessive exposure to ultraviolet light and in particular should avoid sunburn.

BUT IT'S NOT THAT SIMPLE

Although it is important to have a good understanding of the conditions outlined in this chapter and Chapter 3, three important additional components are necessary for successful management of patients with sporting injuries.

Pain: where is it coming from?

The pain your patient feels at a particular site may not necessarily be emanating from that site. It is essential to understand pain, which is the topic of Chapters 5 and 6.

Masquerades

There are many medical conditions whose presentation may mimic a sporting injury. While many of these conditions are relatively rare, it is nevertheless important to keep them at the back of your mind. If the clinical pattern does not seem to fit the obvious diagnosis, then think of the conditions that may masquerade as sporting injuries. These are described in Chapter 7.

The kinetic chain

Every athletic activity involves movements of joints and limbs in coordinated ways to perform a task. These activities include running, jumping, throwing, stopping

or kicking. The tasks may include throwing a ball, hitting a ball, kicking a ball, jumping over an object, or propelling the body through air or water. Individual body segments and joints, collectively called links, must be moved in certain specific sequences to allow efficient accomplishment of the tasks.

The sequencing of the links is called the kinetic chain of an athletic activity.¹⁴² Each kinetic chain has its own sequence but the basic organisation includes proximal to distal sequencing, a proximal base of support or stability, and successive activation of each segment of the link and each successive link. The net result is generation of force and energy in each link, summation of the developed force and energy through each of the links, and efficient transfer of the force and energy to the terminal link.

Injuries or adaptations in some areas of the kinetic chain can cause problems not only locally but distantly, as distal links must compensate for the lack of force and energy delivered through the more proximal links. This phenomenon, called catch-up, is both inefficient in the kinetic chain and dangerous to the distal link because it may cause more load or stress than the link can safely handle. These changes may result in anatomical or biomechanical situations that increase injury risk, perpetuate injury patterns or decrease performance. For example, a tennis player with stiffness of the lumbar spine may overload the rotator cuff muscles while serving to generate sufficient power and, thus, develop a tear of the rotator cuff muscles.

These deficits in the kinetic chain must be identified and corrected as part of the treatment and rehabilitation process. We will be constantly returning to the theme of the kinetic chain throughout the following chapters.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

